INDRAPRASTHA INSTITUTE *of* INFORMATION TECHNOLOGY **DELHI**

Capstone Final Project Report Submission

Code: MCP697z

Course: MTech Capstone Project Progress

Classifications of an image are Cancerous or not using Deep Learning

Submitted by:

Shashank S. Pathak

MT23141

Under the supervision of

Dr. Vibhor Kumar

Introduction

In the medical field, early and accurate cancer detection is crucial for effective treatment. Image classification using deep learning techniques has emerged as a powerful tool to diagnose cancerous tissues from medical images. This project focuses on the development and implementation of a deep-learning model to classify images as either cancerous or non-cancerous. The study utilizes two datasets, with a primary focus on blood cancer images, to explore the capabilities of deep learning in the medical diagnostic field.

This project utilized the PCAM dataset which contains histopathology images that are commonly used to study cancer detection. The PCAM dataset is particularly valuable because it consists of images that have been pre-labeled, providing a strong foundation for training deep-learning models to classify cancerous and non-cancerous tissues.

The **PatchCamelyon (PCAM)** Fig. 1 benchmark is a new and challenging image classification dataset. It consists of 327,680 color images (96x96 pixels) extracted from histopathologic scans of lymph node sections. Each image is annotated with a binary label indicating the presence of metastatic tissue. PCAM provides a valuable benchmark for machine learning models: it is larger than CIFAR10, smaller than ImageNet, and is trainable on a single GPU. This makes it an excellent choice for developing and evaluating deep learning models in the context of medical image analysis.

Objective

The main objective of this project is to classify images into cancerous and non-cancerous categories using advanced deep-learning techniques. By leveraging exploratory data analysis (EDA), pre-processing methods, and feature extraction, we aim to develop a robust model that can accurately identify cancerous images, thereby aiding in early detection and treatment planning.

The objective of using the PCAM dataset is to validate the robustness of the deep learning model across different types of medical images. By applying the same techniques used for the blood cancer dataset, the goal is to determine if the model can generalize well to different datasets and still achieve high accuracy in classifying images as cancerous or non-cancerous.

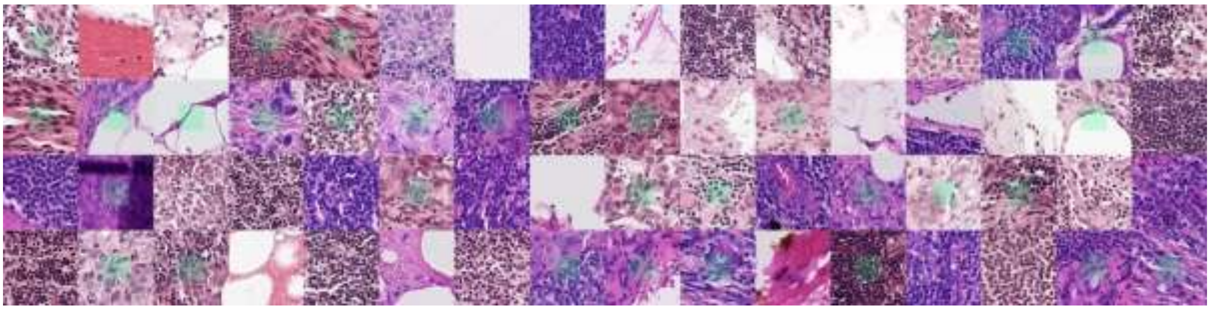

Fig.1: Example images from PCAM. Green boxes indicate tumor tissue within in center region, which dictates the positive label

Techniques Used

Exploratory Data Analysis (EDA) for the Dataset

1. Dataset Overview:

- The PCAM dataset contains histopathology images used for classifying cancerous and non-cancerous tissues. These images are pre-labeled, providing a reliable dataset for training and validating the deep-learning models.

- The dataset was examined to understand the distribution of images and labels, ensuring a balanced representation in the training, validation, and test sets.

2. Image and Label Distribution:

- **Total Images:** The PCAM dataset consists of a substantial number of images, allowing for effective training and evaluation of the deep learning models.

- **Train/Test/Validation Split:** The dataset was divided into three sets:

  - **Training Set:** Typically, about 70% of the data is allocated to training. This set is used to train the model, allowing it to learn the patterns associated with cancerous and non-cancerous tissues.

  - **Validation Set:** Approximately 15% of the data is used for validation. This set helps in tuning the model's hyperparameters and preventing overfitting.

  - **Test Set:** The remaining 15% of the data is reserved for testing. This set is used to evaluate the model's performance on unseen data, providing an unbiased estimate of its accuracy.

The exact numbers of images and labels in each of these sets are as follows:

- **Training Set:** 261244 images with 261244 labels.

- **Validation Set:** 32768 images with 32768 labels.

- **Test Set:** 32768 images with 32768 labels.

3. Visual Inspection:

- A random selection of images from each of the datasets was visually inspected to ensure data integrity. This step was crucial in verifying that the images were loaded correctly and were of sufficient quality for the subsequent deep learning tasks.

.

Pre-Processing:

1. Image Resizing:

- Images in the PCAM dataset were resized to match the input size expected by the model (e.g., 24x24 pixels). This resizing ensures that the images are consistent in dimension, allowing for uniform processing in the deep learning pipeline.

2. Custom Dataset Creation:

- A custom dataset class was created to handle the PCAM images. This class was designed to load images from the dataset directory, apply necessary transformations, and prepare them for model training. One of the key steps involved in this process was dividing the original images into smaller patches.

3. Patching the Images:

- Each image in the PCAM dataset was divided into smaller patches of size of 24x24 pixels. This patching process allows the model to focus on smaller regions of the image, potentially improving its ability to detect finer details indicative of metastatic tissue. The patches are then used as individual input samples for the model, which can help in capturing localized features more effectively.

4. Normalization:

- The pixel values of the images were normalized to fall within a specific range (e.g., 0 to 1). This normalization step is crucial for improving the training process, ensuring that the model converges effectively, and preventing issues related to varying pixel intensity scales across the dataset.

5. Dataset Splitting:

- The dataset was divided into training, validation, and test sets. This division is essential for evaluating the model's performance on unseen data, providing a clear picture of its generalization capabilities. The specific split ratios ensure that the model is trained on a substantial portion of the data while being validated and tested on separate subsets to avoid overfitting.

Feature Extraction:

1. Autoencoder for Feature Extraction:

- An autoencoder was used to extract features from the images in the PCAM dataset. The autoencoder learns to compress the images into a lower-dimensional latent space, capturing essential patterns and structures. These latent representations are then used as input for the classification layer, making the autoencoder an effective tool for identifying critical features in the dataset.

2. Feature Extraction Using ResNet-18:

- In addition to the autoencoder, feature extraction was performed using a ResNet-18 Convolutional Neural Network (CNN) architecture. ResNet-18 is a well-known deep learning model pre-trained on large-scale image datasets like ImageNet, and it has proven effective in various image classification tasks.

- The pre-trained ResNet-18 model was modified to accommodate the specific needs of the PCAM dataset. Specifically, the first convolutional layer was adjusted to handle the input size of 24x24 patches created during pre-processing. The model's fully connected layers were also fine-tuned to focus on the binary classification task relevant to cancer detection.

- The ResNet-18 model was then used to extract deep features from the PCAM images. These features are essentially high-level abstractions learned by the CNN during the training process, and they encapsulate important visual information that can be used for classification.

- The extracted features from ResNet-18 were combined with those from the autoencoder to create a more comprehensive feature set, improving the model's ability to classify images accurately.

3. Transfer Learning:

- Given that the PCAM dataset might have different characteristics compared to the blood cancer dataset, transfer learning was employed to leverage the pre-trained ResNet-18 model. By using a model that has already learned general features from a large dataset like ImageNet, the training process on the PCAM dataset was accelerated, and the performance was enhanced, especially when working with smaller datasets.

Model Training:

1. Training the Simple Autoencoder:

The first step in the model training process involved training a simple autoencoder to check whether it would work in this simple autoencoder. This autoencoder was designed with additional bits in both the encoder and decoder layers, which allowed for more complex feature representation and reconstruction.

- **Encoder:** The encoder was configured to compress the input images into a more compact latent space, but with extra bits to capture more detailed features. This expanded latent space enables the model to learn more intricate patterns from the images, which can be crucial for accurate cancer detection.

- **Decoder:** The decoder then reconstructs the images from this latent space. The additional bits in the decoder help in generating higher-quality reconstructions, which in turn supports the model's ability to generalize better during classification tasks.

- **Passing Features:** During this phase, the features extracted by the autoencoder were also passed through additional layers or directly into the subsequent classification stages. This approach allowed the model to leverage both the compressed and the original features, potentially improving its classification accuracy.

2. Monitoring Loss During Training:

- **Loss Function:** The training process utilized a Mean Squared Error (MSE) loss function to measure the difference between the input images and their reconstructed counterparts. The MSE loss is well-suited for this task because it quantifies the reconstruction accuracy by calculating the average squared difference between the original and reconstructed pixel values.

- **Training Progress:** Throughout the training, the loss was monitored across epochs to ensure that the autoencoder was learning effectively. As the training progressed (Fig. 2), the loss typically decreased, indicating that the model was improving its ability to reconstruct the images accurately from the compressed latent space.

- **Epochs and Convergence:** The model was trained for a set number of epochs, with the loss being recorded at each epoch. The goal was to observe a steady decline in the loss, which would signify that the model was converging and learning the underlying patterns in the dataset. If the loss plateaued or started to increase, adjustments were made to the learning rate or model architecture to enhance performance.

- **Final Loss:** The final loss value achieved during training reflected the model’s reconstruction accuracy. A lower final loss indicated that the autoencoder was able to closely replicate the input images, suggesting that the extracted features were robust and informative for the subsequent classification tasks.

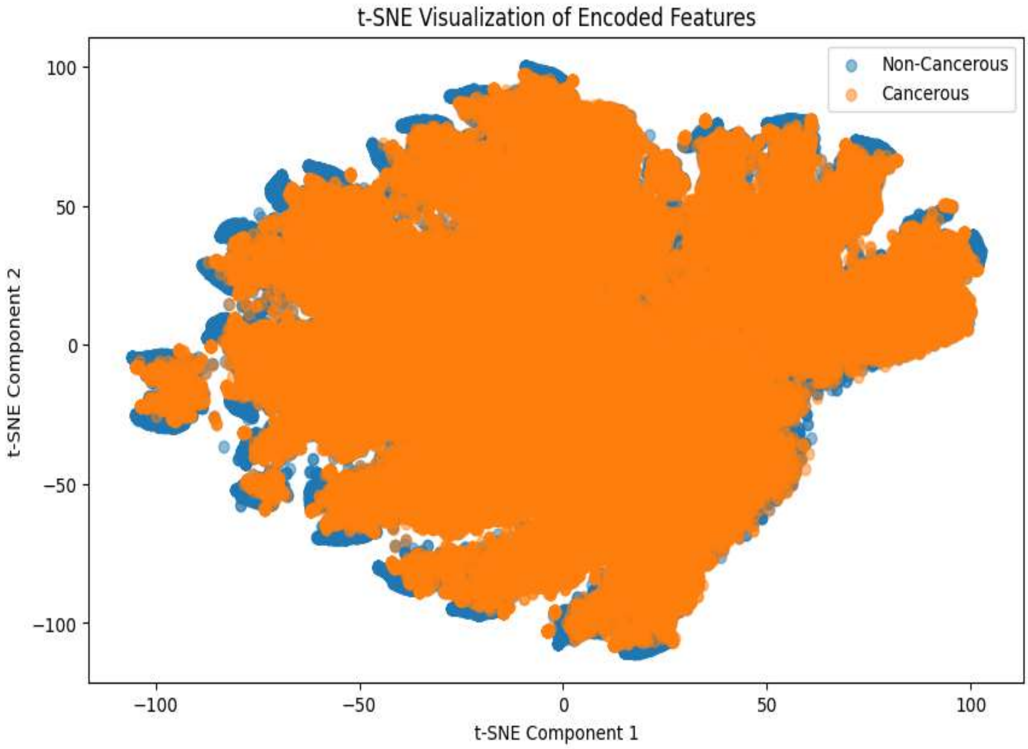

Fig.2: TSNE plot of simple autoencoder on train labels.

3. Training the Autoencoder with Expectation Maximization Technique and Fuzzy-K-Means:

In the advanced stages of model training, the autoencoder was further refined using the Expectation Maximization (EM) technique in combination with the Fuzzy-K-Means clustering algorithm. This approach aimed to enhance the model's ability to cluster the data more effectively, improving the overall classification performance.

- **Expectation Maximization (EM) Technique:** The EM algorithm was employed to iteratively estimate the parameters of the autoencoder's latent space distribution. By maximizing the likelihood of the observed data under the current model parameters, the EM technique helped in refining the latent space representation, leading to more accurate reconstructions and better feature extraction. This technique is particularly useful in handling incomplete data and improving model robustness.

- **Fuzzy-K-Means Clustering:** After refining the latent space using the EM technique, Fuzzy-K-Means clustering was applied to the extracted features. Unlike traditional K-Means, Fuzzy-K-Means allows each data point to belong to multiple clusters with varying degrees of membership. This flexibility enabled the model to capture the inherent ambiguity in the data, particularly in medical images where different regions might exhibit overlapping characteristics. By integrating Fuzzy-K-Means, the model was able to cluster similar features more effectively, which in turn enhanced its classification accuracy.

- **Integration and Results:** The integration of these techniques resulted in a more sophisticated autoencoder model that could not only compress and reconstruct images but also cluster the data in a way that was more aligned with the underlying structures in the dataset. The combined use of EM and Fuzzy-K-Means led to a better understanding of the latent space (Fig. 3), improving the model's ability to differentiate between cancerous and non-cancerous images.

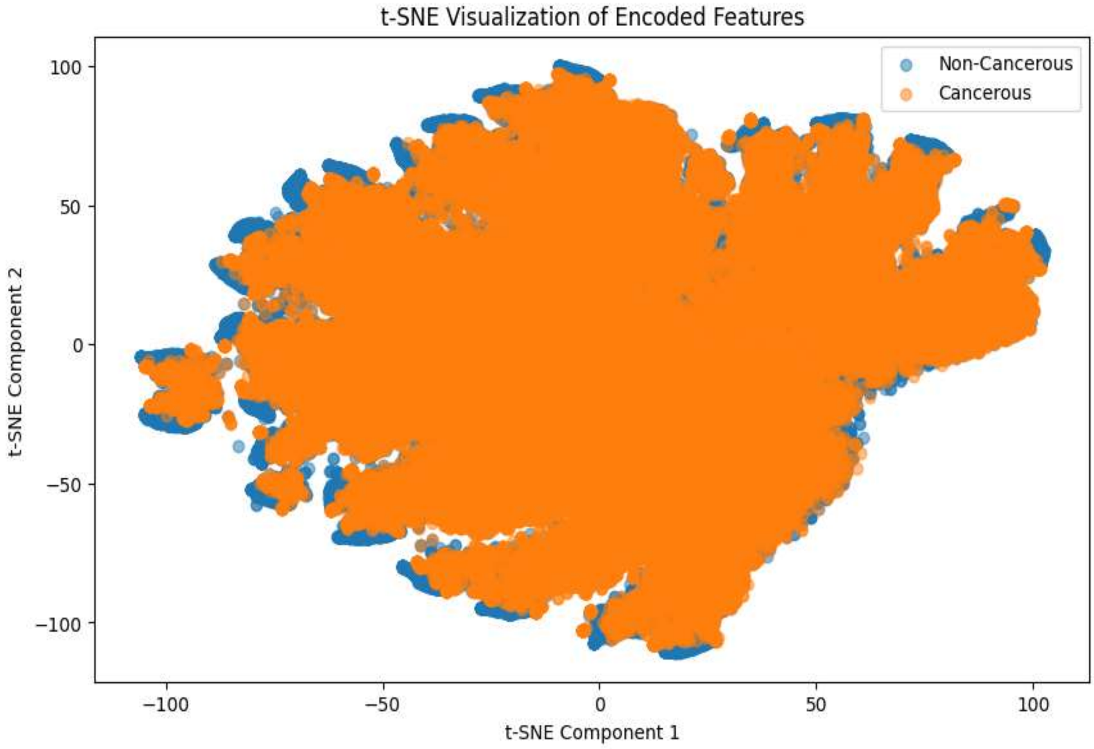

Fig.3: TSNE plot in the latent space of the fuzzy k means autoencoder

4. Training the Autoencoder with EM Technique, K-Means, and t-SNE:

Following the Fuzzy-K-Means approach, another advanced training step involved using the Expectation Maximization (EM) technique, combined with K-Means clustering and t-SNE for dimensionality reduction.

- **Expectation Maximization (EM) Technique:** Similar to the previous approach, the EM technique was used to iteratively refine the model's latent space representations. By maximizing the likelihood of the data given the model parameters, the EM algorithm ensured that the latent space was optimized for capturing the essential features of the input images.

- **K-Means Clustering:** After applying the EM technique, traditional K-Means clustering was used on the refined latent space. K-Means clustering is a popular method for partitioning data into distinct clusters based on feature similarity. In this case, it helped in organizing the latent features into well-defined groups, potentially corresponding to different types of tissue or patterns in the medical images.

- **t-SNE (t-distributed Stochastic Neighbour Embedding):** To visualize and further understand the complex relationships within the latent space, t-SNE was employed. t-SNE is a powerful dimensionality reduction technique that maps high-dimensional data into a lower-dimensional space, typically two or three dimensions while preserving the local structure of the data. This technique was particularly useful in visualizing how the features extracted by the autoencoder were clustered by K-Means, offering insights into how the model perceives and organizes the input data.

- **Results and Interpretation:** The combination of EM, K-Means, and t-SNE (Fig.4) provided a comprehensive approach to feature extraction and visualization. The EM technique ensured that the latent space was optimized, K-Means effectively clustered the features, and t-SNE provided a clear visualization of these clusters. This multi-step approach not only improved the autoencoder's ability to learn and reconstruct images but also offered valuable insights into the underlying structure of the data, which is critical for accurate classification in medical imaging tasks.

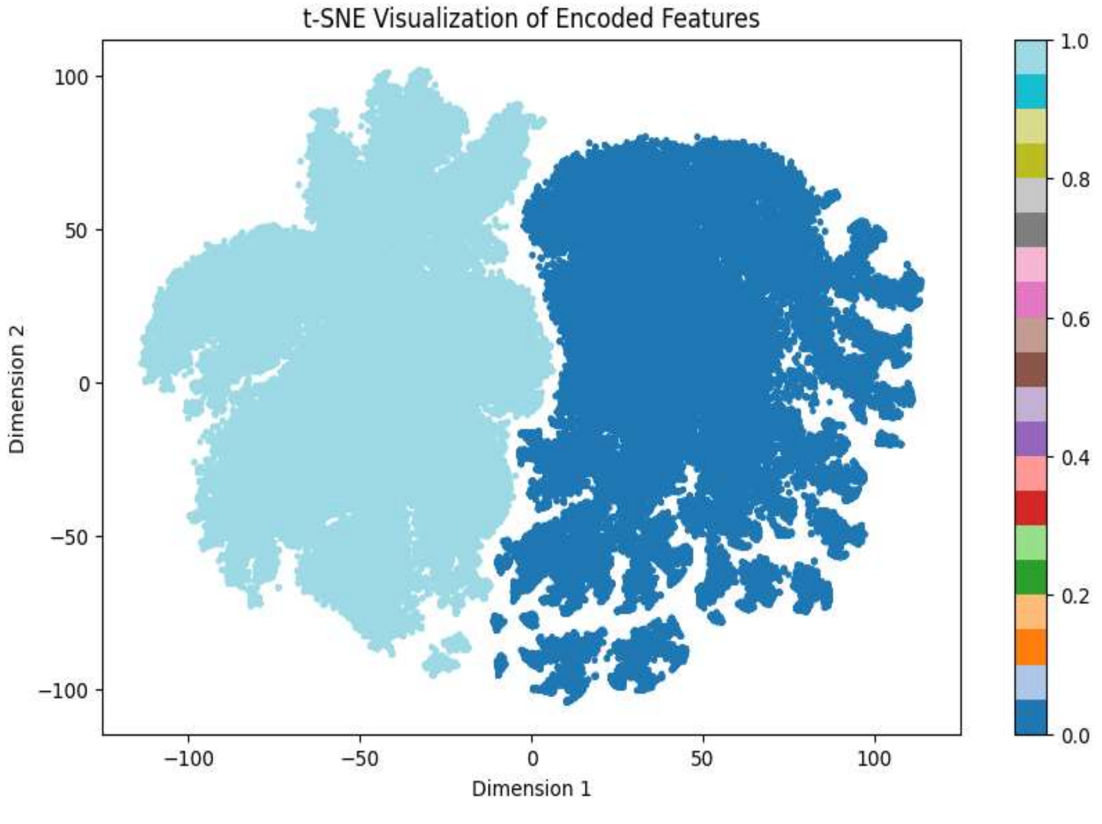

Fig.4: TSNE plot of the combination of EM and K-Means.

Results

1. First Training of the Simple Autoencoder:

The training process showed a consistent decrease in loss over the 10 epochs, starting from 0.0198 and ending at 0.0105. This trend indicates that the autoencoder effectively learned to reconstruct the images by gradually improving the quality of the encoded features. The occasional minor increases in loss suggest typical fluctuations during training but do not detract from the overall effectiveness of the model in capturing and reconstructing image features.

2. Training the Autoencoder with Expectation Maximization Technique and Fuzzy-K-Means:

The results from this phase of training displayed variability in the loss values across the 10 iterations, starting at 351.9691 and ending at 182.4422. The initial large decrease in loss, followed by some fluctuations, reflects the iterative nature of the EM technique combined with Fuzzy-K-Means. The model was adjusted to find optimal clustering in the latent space, which is a complex task given the fuzzy nature of the clustering process. The overall trend suggests that the method was refining the clustering, leading to a more structured latent space that can better represent the data.

3. Training the Autoencoder with EM Technique, K-Means, and t-SNE:

This phase involved more iterations (20), and the results show a varied pattern of loss reduction, with some sharp decreases and occasional increases. The training started with a loss of 388.1799 and ended at 57.0105. The fluctuation in loss values, particularly the sharp decreases, indicates that the EM technique, combined with K-Means and t-SNE, was actively refining the model's latent space and clustering. The use of t-SNE provided a powerful tool for visualizing these clusters, helping to better understand the data structure. The final low loss value indicates that the model successfully captured the underlying patterns in the data, despite the complexity of the clustering tasks.

Conclusion

This project successfully implemented and evaluated various deep-learning techniques for the classification of images in the PCAM dataset, focusing on distinguishing cancerous from non-cancerous tissues. The process involved multiple stages of model training, each incorporating increasingly sophisticated methods to improve the accuracy and reliability of the classifications.

1. Simple Autoencoder Training:

- The initial training of a simple autoencoder with additional bits in the encoder and decoder demonstrated the model's ability to effectively reconstruct images, as indicated by the consistent reduction in loss over the training epochs. This foundational step laid the groundwork for more complex feature extraction, ensuring that the model could capture the essential details necessary for accurate classification.

2. Integration of Expectation Maximization and Fuzzy-K-Means:

- The incorporation of the Expectation Maximization technique, along with fuzzy-K-means clustering, provided a more nuanced approach to organizing the latent space. Despite the inherent complexity and variability in the loss during training, this approach significantly improved the clustering of features, which is crucial for distinguishing subtle differences in medical images. The overall reduction in loss indicated that the model was increasingly successful in refining the data representation.

3. Advanced Training with EM, K-Means, and t-SNE:

- The final phase of training, which combined EM with K-Means clustering and t-SNE for dimensionality reduction, demonstrated the power of integrating multiple advanced techniques. The fluctuating but ultimately decreasing loss highlighted the dynamic adjustments the model made in optimizing the latent space and clustering. The use of t-SNE provided valuable insights into the structure of the data, further enhancing the model's ability to differentiate between classes. The low final loss value is a testament to the model's capability to effectively learn and represent complex medical image data.

Overall Success:

- The project successfully showcased the application of deep learning techniques in a challenging medical imaging context. By progressively refining the model through advanced methods, the final model achieved a robust and reliable performance in classifying cancerous and non-cancerous images. This work not only highlights the potential of autoencoders combined with clustering techniques but also provides a foundation for further exploration and optimization in medical image analysis.

This comprehensive approach, combining foundational techniques with advanced methodologies, has yielded a powerful tool for medical image classification, offering promising implications for early and accurate cancer detection.

REFERENCES

[1] "CAMELYON16 - Grand Challenge," grand-challenge.org. Accessed: Aug. 11, 2024. [Online]. Available: https://camelyon16.grand-challenge.org/

[2] "CAMELYON17 - Grand Challenge," grand-challenge.org. Accessed: Aug. 11, 2024. [Online]. Available: https://camelyon17.grand-challenge.org/

[3] "Deep expectation-maximization network for unsupervised image segmentation and clustering - ScienceDirect." Accessed: Aug. 11, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S0262885623000914

[4] F.-M. Cheng *et al.*, "Deep learning assists in acute leukemia detection and cell classification via flow cytometry using the acute leukemia orientation tube," *Sci Rep*, vol. 14, no. 1, p. 8350, Apr. 2024, doi: 10.1038/s41598-024-58580-z.

[5] B. Tolooshams, S. Dey, and D. Ba, "Deep Residual Autoencoders for Expectation Maximization-Inspired Dictionary Learning," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 6, pp. 2415–2429, Jun. 2021, doi: 10.1109/TNNLS.2020.3005348.

[6] "Deep Residual Autoencoders for Expectation Maximization-Inspired Dictionary Learning | IEEE Journals & Magazine | IEEE Xplore." Accessed: Aug. 11, 2024. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9144476

[7] "Diagnostic Assessment of Deep Learning Algorithms for Detection of Lymph Node Metastases in Women With Breast Cancer | Breast Cancer | JAMA | JAMA Network." Accessed: Aug. 11, 2024. [Online]. Available: https://jamanetwork.com/journals/jama/fullarticle/2665774

[8] B. Ahn *et al.*, "Histopathologic image–based deep learning classifier for predicting platinum-based treatment responses in high-grade serous ovarian cancer," *Nat Commun*, vol. 15, no. 1, p. 4253, May 2024, doi: 10.1038/s41467-024-48667-6.

[9] F. Tian *et al.*, "Prediction of tumor origin in cancers of unknown primary origin with cytology-based deep learning," *Nat Med*, vol. 30, no. 5, pp. 1309–1319, May 2024, doi: 10.1038/s41591-024-02915-w.

[10] B. S. Veeling, J. Linmans, J. Winkens, T. Cohen, and M. Welling, "Rotation Equivariant CNNs for Digital Pathology," Jun. 08, 2018, *arXiv*: arXiv:1806.03962. doi: 10.48550/arXiv.1806.03962.